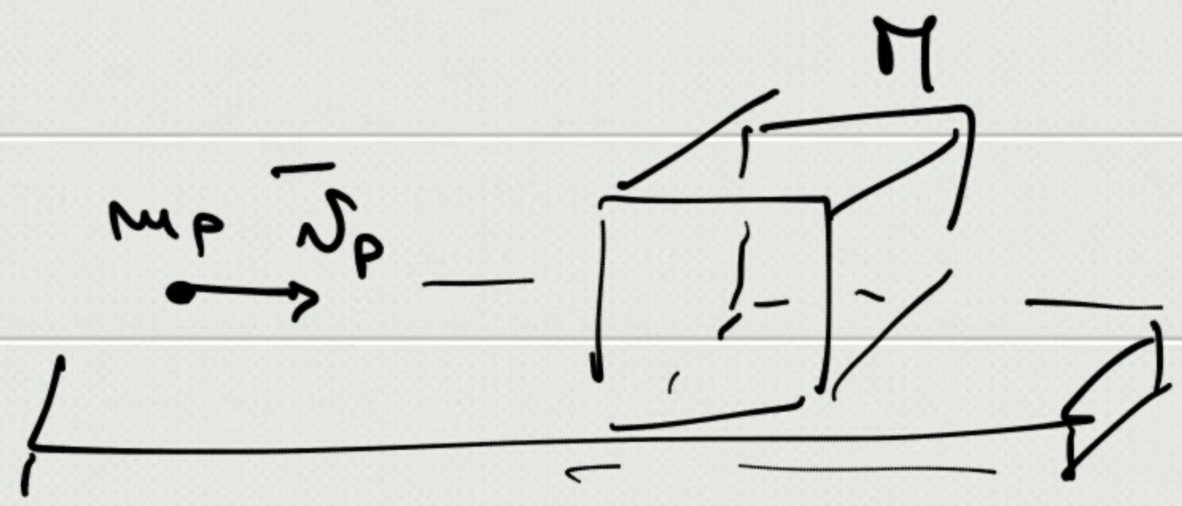
$m_p = 0.05\ kg$

$T_p = 293\ K$



$v_p = 100\ m/s$ $\qquad M = 0.5\ kg$ $\qquad T_g = 273.15\ K$

$c_p = 130\ J/kg\,K$ $\qquad \lambda_g = 3.3 \cdot 10^5\ J/kg$ $\qquad m_g = ?$ (fonde)

$$\bar{P} = cost \Rightarrow m_p v_p = (m_p + M) V \Rightarrow V = \frac{m_p}{m_p + M} v_p$$

$$E_{diss} = E_{k,i} - E_{kf} = \frac{1}{2} m_p v_p^2 - \frac{1}{2}(m_p + M) V^2 = 227\ J$$

$\hookrightarrow Q$

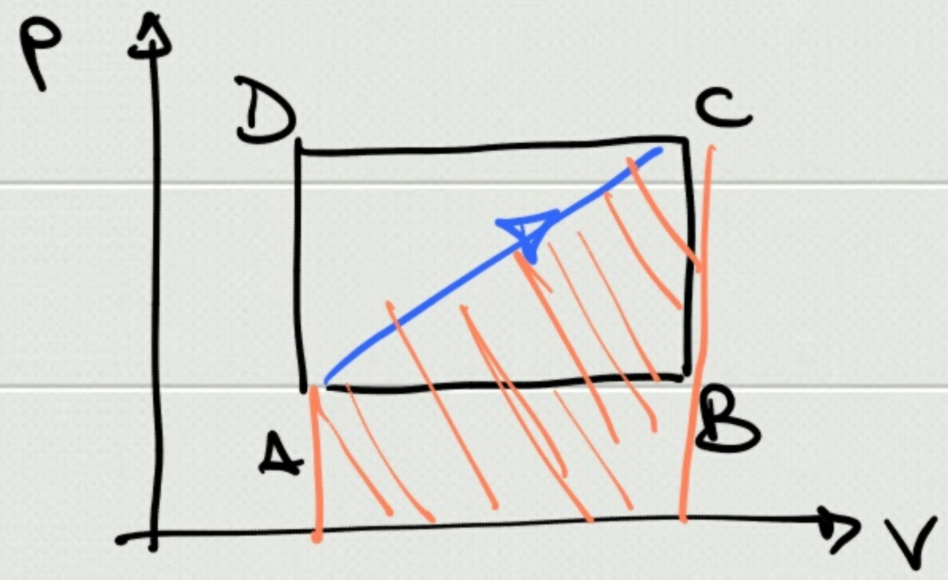
$$E_{diss} = m_g \lambda_g$$

$$= m_g c_g (T_g - T_p)$$

$$= m_g \lambda_g + m_p c_p (T_g - T_p) \quad *$$

$$= m_g c_g (T_g - T_p) + m_p c_p (T_g - T_p)$$

$$\Rightarrow m_g = 1.1 \cdot 10^{-3}\ kg$$



$W_{ABC} = 50\ J$

$Q_{ABC} = 83{,}5\ J$

$W_{ciclo} = -40\ J$

ΔU_{AC}, Q_{CDA}, W_{AC}, Q_{AC}

$$\Delta U_{ABC} \left(= \Delta U_{AC}\right) = Q_{ABC} - W_{ABC} = 33.5\ J$$

$$\Delta U_{ciclo} = 0 \quad \overset{1°P.}{\Longrightarrow} \quad Q_{ciclo} = W_{ciclo}$$

$$Q_{ciclo} = Q_{ABC} + Q_{CDA} \Rightarrow Q_{CDA} = W_{ciclo} - Q_{ABC} =$$

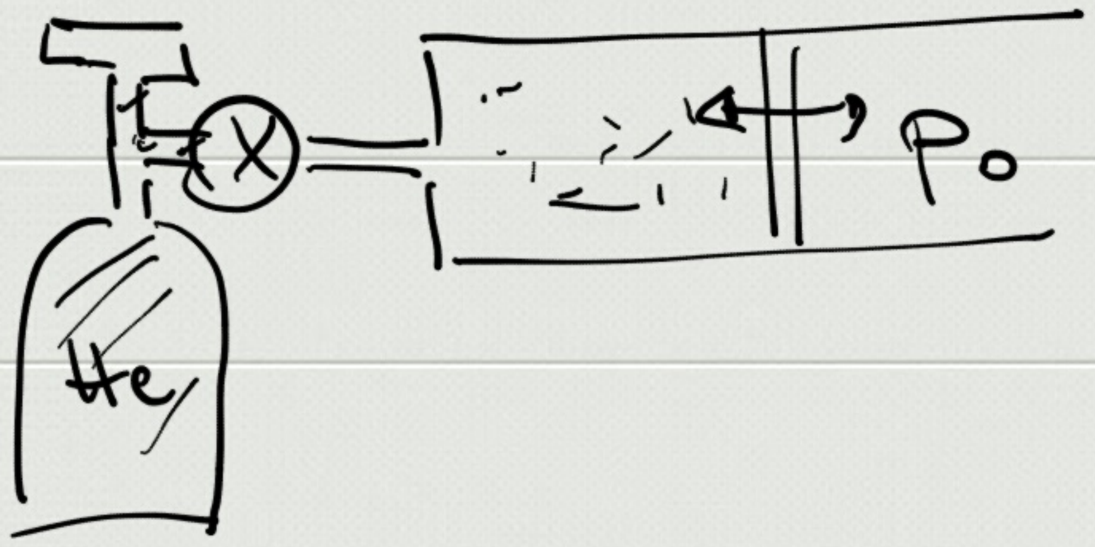
$$= -123.5\ J$$

$$W_{AC} = W_{ciclo} - W_{ABC}$$

$$\overset{!}{=} W_{ABC} + \frac{1}{2} W_{ciclo}$$

$$\overset{!}{=} -\Delta U_{AC}$$

$$\overset{!}{=} W_{ABC} + \frac{1}{2} |W_{ciclo}| \ * \ = 103.5\ J$$



$n = 4 \qquad T = 300\ K \qquad p = 5 \cdot 10^6\ Pa$

$p_0 = 1.013 \cdot 10^5\ Pa = cost$

$V_{g,0} = 0 \qquad T = cost$ (isoterma)

$W_{gas,\, 0 \to eq.} \qquad Q_{ced,\, amb}$

$$W = \int p\, dV = p_0\, \Delta V = p_0 (V_f - V_i)$$

$$p V_i = nRT \Rightarrow V_i = \frac{nRT}{p} = 0.0002\ m^3$$

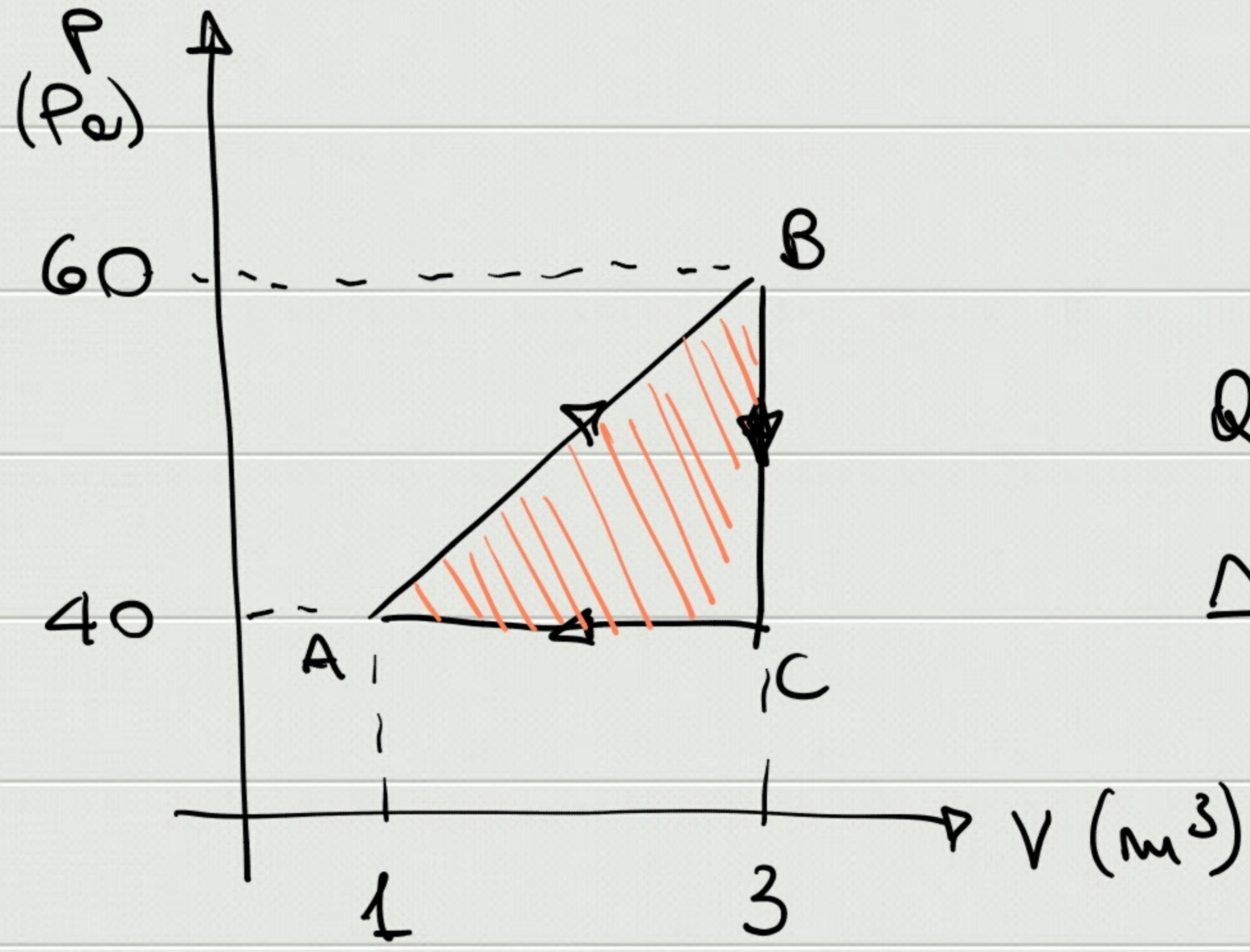
$$p_0 V_f = nRT \Rightarrow V_f = \frac{nRT}{p_0} = 0.0985\ m^3$$

$$\Rightarrow W = 9.775 \cdot 10^3\ J$$

gas : $\Delta U = 0 \Rightarrow Q_{gas} = W_{gas} =$

$= 9.775 \cdot 10^3 \ J$

$Q_{amb} = - Q_{gas} = - 9.775 \cdot 10^3 \ J$



$Q_{ciclo} = ?$

$\Delta U_{ciclo} = 0$

$$\Rightarrow Q_{ciclo} = W_{gas}$$

$$Q_{ciclo} = \frac{1}{2}(p_B - p_A)(V_C - V_A) \quad *$$

$$\overset{!}{=} \frac{1}{2}(p_A + p_B)(V_C - V_A)$$

$$\overset{!}{=} \frac{1}{2}(p_B - p_A)(V_A + V_C)$$

$$\overset{!}{=} \frac{1}{2}(p_A + p_B)(V_C - V_A) + mc_V(T_B - T_A) +$$

$$\underbrace{+ mc_V(T_C - T_B) + mc_p(T_A - T_C)}_{\underbrace{(W_{AB} + \Delta U_{AB})}_{Q_{AB}}} + Q_{BC} + Q_{CA}$$